

Real Estate DI2 - Shadow Listing Audit Example

Purpose

This example shows how Real Estate DI2 converts a listing that feels overpriced into a structured, auditable buy/no-buy posture.

The point is not to decide whether the house is pretty. The point is to determine whether the deal logic holds.

I. Snapshot

Item	Value
Market	West Chester-area / Southeastern Pennsylvania
Approximate Listing Price	Approximately \$625K
Bedrooms	4
Bathrooms	1 full + 2 half
Lot	Approximately 0.5 acre
Garage	1-car
Year Built	Late 1960s
HOA	No
Status	Coming Soon
Market Condition	Hot / constrained inventory

II. Gate Results

Because the property is listed as Coming Soon, not verified Active, final execution is blocked.

Gate	Result
Active Listing Gate	Blocked - Coming Soon is not verified Active
Source Reliability	Provisionally acceptable
Lag / Status Risk	Elevated because status may change
Final Execution	Blocked until Active status is verified

Shadow analysis may run, but offer execution remains blocked until the listing is verified Active.

III. Formula Layer

The formula layer is what turns a normal house vibe check into an auditable decision posture.

The formulas below are shown in simplified public form. They are not appraisal formulas, lending formulas, or investment advice. They are internal decision-support checks used to structure risk, confidence, and execution discipline.

A. Active Listing Gate

Purpose: Prevent final execution when the property is not yet verified Active.

```
VASG = 1 if L_active = 1 and SRI x LFS >= threshold
VASG = 0 otherwise
```

Variable	Meaning
VASG	Verified Active Status Gate
L_active	Whether listing is verified Active
SRI	Source Reliability Index
LFS	Listing Freshness Score
threshold	Required execution confidence level

```
Coming Soon != verified Active
L_active = 0
VASG = 0
```

Result: final execution blocked.

B. Listing Freshness Score

Purpose: Account for status lag and stale-listing risk.

```
LFS = (1 - ELRF_hat) x ESR_hat
```

Variable	Meaning
LFS	Listing Freshness Score
ELRF_hat	Estimated lag / status-risk factor
ESR_hat	Estimated source / audit-seal confidence

```
LFS = (1 - 0.25) x 0.80
LFS = 0.60
```

Result: freshness is not strong enough to override Coming Soon status.

C. Condition Sensitivity Check

Purpose: Penalize the listing for known feature or condition risks.

```
CSCS = 1 - (RiskWeight x ConditionPenalty)
```

Variable	Meaning
CSCS	Condition Sensitivity Check Score
RiskWeight	Weight assigned to condition risk
ConditionPenalty	Estimated condition / feature penalty

```
CSCS = 1 - (0.15 x 0.20)
CSCS = 0.970
```

Result: condition concerns exist, but they do not destroy the audit posture by themselves.

D. Composite Suitability Index

Purpose: Combine major feature limitations into a single suitability score.

```
CSI = (1 - BathroomPenalty) x (1 - GaragePenalty) x (1 - Age/LayoutPenalty)
```

Variable	Meaning
CSI	Composite Suitability Index
BathroomPenalty	Penalty for bathroom count / layout
GaragePenalty	Penalty for garage limitation
Age/LayoutPenalty	Penalty for older build, dated layout, or modernization drag

$$CSI = (1 - 0.145) \times (1 - 0.15) \times (1 - 0.12)$$

$$CSI = 0.639$$

Result: the house has real feature drag. The main drag is not price alone; it is price relative to limitations.

E. Value Confidence Index

Purpose: Combine source confidence with property suitability.

$$VCI = ESR_hat \times CSI$$

Variable	Meaning
VCI	Value Confidence Index
ESR_hat	Estimated source / audit confidence
CSI	Composite Suitability Index

$$VCI = 0.80 \times 0.639$$

$$VCI = 0.511$$

Result: value confidence is moderate, not clean. The property is audit-worthy, but not a chase asset.

F. Price Efficiency Ratio - Adjusted

Purpose: Compare estimated fair-value anchor to asking posture, then adjust for feature risk.

$$PER_adj = (FairValueAnchor / ApproximateAsk) \times (1 - FeaturePenalty)$$

Variable	Meaning
PER_adj	Adjusted Price Efficiency Ratio
FairValueAnchor	Internal rounded fair-value anchor
ApproximateAsk	Rounded asking posture
FeaturePenalty	Penalty for layout, bathroom, garage, or condition limitations

$$PER_adj \sim (high-\$500Ks / low-to-mid-\$600Ks) \times (1 - feature\ penalty)$$

$$PER_adj \sim 0.88$$

Result: the price is not irrational, but efficiency is below ideal.

G. Price Plausibility Score

Purpose: Measure whether the listing price is within a defensible market corridor.

$$PPS = 1 - OVI_hat$$

Variable	Meaning
PPS	Price Plausibility Score
OVI_hat	Estimated overvaluation index

$$PPS = 1 - 0.10$$

$$PPS = 0.900$$

Result: list posture is market-plausible.

H. Appraisal / Market Support Proxy

Purpose: Estimate whether comparable-market support exists.

$$APP = \text{sigmoid-style market-support function}$$

Input	Effect
Comparable sales support the price band	APP increases

Input	Effect
Market inventory is tight	APP increases
Location premium is credible	APP increases
Lot size supports value	APP increases
Feature drag is high	APP decreases
Price is detached from comps	APP decreases
Condition risk is unresolved	APP decreases

APP $\approx$ 0.81

Result: market support appears meaningful but not decisive.

I. Friction / Missing Support Gate

Purpose: Block execution when key property facts are unverified.

FMS = 1 only if first-order property risks are verified or cleared
FMS = 0 if material risks remain unresolved

Material risks include	roof age, HVAC age, water heater age, basement moisture, electrical panel, plumbing, deck condition, drainage, inspection status
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FMS = 0

Result: execution blocked pending verification.

J. Hazard / Unknowns Gate

Purpose: Block final posture when material hazards remain unknown.

HAU = 1 if hazard audit is sufficiently complete
HAU = 0 if hazards remain unresolved

Hazards include	water intrusion, structural deck problems, major mechanical defects, old electrical systems, drainage issues, environmental or site-specific concerns
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HAU = 0

Result: final offer execution blocked.

K. Internal Delay / Timing Risk

Purpose: Measure timing pressure caused by Coming Soon status or market velocity.

IDRI = DelayRisk x MarketHeat x StatusUncertainty

Variable	Meaning
IDRI	Internal Delay / Timing Risk Index

IDRI $\approx$ 0.29

Result: timing risk exists, but should not override status gates.

L. Offer Fatigue / Emotional Surge Check

Purpose: Prevent the buyer from chasing because of scarcity or frustration.

OFES = BuyerFatigue x MarketPressure x EmotionalPremiumRisk

Variable	Meaning
OFES	Offer Fatigue / Emotional Surge score

OFES $\approx$ 0.33

Result: buyer-emotion risk exists and must be capped with a walk-away number.

M. Ethical / Inversion Gate

Purpose: Detect whether the offer logic has inverted from disciplined buying to panic buying.

$$\text{EIG} = 1 - \text{IRTR}$$

Variable	Meaning
EIG	Ethical / Inversion Gate
IRTR	Inversion Risk Trigger Rate

$$\begin{aligned}\text{IRTR} &= 0.00 \\ \text{EIG} &= 1.00\end{aligned}$$

Result: no ethical or logic inversion detected at the shadow-analysis stage.

N. Global Offer Hazard

Purpose: Combine execution risks into a caution score.

GOH = weighted composite of listing status uncertainty, feature drag, market heat, inspection unknowns, and emotional overbid risk

Variable	Meaning
GOH	Global Offer Hazard

$$\text{GOH} \approx 0.41$$

Result: moderate offer hazard.

O. Offer Permission Score

Purpose: Determine whether the system may move from shadow analysis to executable offer posture.

$\text{OPS_shadow} = \text{function}(\text{PPS}, \text{APP}, \text{VCI}, \text{CSCS}, \text{EIG}, \text{GOH}, \text{FMS}, \text{HAU})$
 $\text{OPS_executable} = \text{VASG} \times \text{OPS_shadow}$

Variable	Meaning
OPS_shadow	Shadow offer permission score
OPS_executable	Executable offer permission score

$$\begin{aligned}\text{VASG} &= 0 \\ \text{FMS} &= 0 \\ \text{HAU} &= 0\end{aligned}$$

$$\begin{aligned}\text{OPS_shadow} &= \text{blocked} \\ \text{OPS_executable} &= 0\end{aligned}$$

Result: no executable offer posture until the listing is Active and material facts are verified.

IV. Formula Summary Table

Formula	Public Meaning	Example Result
VASG	Active-listing execution gate	0 / blocked
LFS	Listing freshness and lag confidence	0.60
CSCS	Condition sensitivity check	0.970
CSI	Composite suitability after feature penalties	0.639
VCI	Value confidence after suitability adjustment	0.511
PER_adj	Adjusted price efficiency	~0.88

Formula	Public Meaning	Example Result
PPS	Price plausibility	0.900
APP	Market support proxy	~0.81
FMS	Material facts verification gate	0 / blocked
HAU	Hazard unknowns gate	0 / blocked
IDRI	Timing / delay risk	~0.29
OFES	Emotional overbid / fatigue risk	~0.33
EIG	Inversion guard	1.00
GOH	Global offer hazard	~0.41
OPS	Final offer permission score	0 / blocked

V. Market Math

The emotional reaction:

This looks like a high-\$500Ks house to a 2024 buyer brain.

The market math:

The low-to-mid \$600Ks list posture may be defensible in the current West Chester-area market if comps, lot, and condition support it.

Lens	Value Read
2024 buyer brain	High \$500Ks
Condition-adjusted fair value	High \$500Ks to low \$600Ks
Current hot-market listing logic	Low-to-mid \$600Ks
Emotional overbid zone	Mid-\$600Ks and above

The gap is the point. The buyer's instinct may be rational based on the physical house, but the market may still support a higher number because of location, lot size, constrained inventory, and comparable sales.

VI. Feature Audit

Feature	Effect
4 bedrooms	Supports value
Approximately half-acre lot	Supports value
West Chester-area market	Supports value
Older late-1960s build	Requires inspection discipline
1 full bath	Major ceiling / resale limitation
1-car garage	Secondary limitation
Dated interior	Cosmetic and renovation discount
Deck / exterior systems	Inspection risk
Unknown roof / HVAC / water heater age	Due diligence risk

VII. Risk Flags

- Bathroom count is the primary valuation ceiling.
- One-car garage limits premium positioning.
- Older construction requires inspection discipline.
- Dated interiors may hide deferred maintenance.
- Deck, roof, HVAC, electrical, plumbing, drainage, and basement moisture should be reviewed.
- The property may be market-defensible without being emotionally satisfying.
- Formula output supports audit-worthiness, not offer execution.

VIII. Offer Logic

Tier	Number
Fair value corridor	High \$500Ks to low \$600Ks
Strong but disciplined offer	Low-to-mid \$600Ks
Max rational	Mid-\$600Ks
Emotional max	Around \$650K
Walk-away	Mid-\$600Ks and above
Inversion zone	Upper-\$600Ks

The list posture is defensible, but the bathroom count and garage ceiling prevent this from being a clean chase-above-market asset.

IX. Final DI2 Read

Real Estate DI2 does not decide whether the house is pretty. It decides whether the deal logic holds.

Question	Answer
Is the property audit-worthy?	Yes
Is the list posture obviously irrational?	No
Is the house physically compelling enough to chase?	No
Is the market premium explainable?	Yes
Should execution proceed before Active verification?	No
Final posture	Audit-worthy, but do not chase

X. Product Takeaway

This example shows why Real Estate DI2 is useful. A buyer may emotionally anchor to an older price expectation, while the current market supports a higher number. The system separates emotional sticker shock from comp-supported value, then applies formula gates, feature penalties, inspection risks, status controls, and offer ceilings.

The result is not a prediction. It is a disciplined decision posture.

XI. Closing Line

The buyer's brain says high \$500Ks. The market says low-to-mid \$600Ks. Real Estate DI2 audits the gap.

Disclaimer

This artifact is for demonstration and internal decision-support purposes only. It is not financial, legal, appraisal, mortgage, or investment advice. All property decisions require independent verification, professional inspection, and qualified real estate guidance.